STAT 335 - Week 1 Times Series R Practice

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```r
#install.packages("astsa")
library("astsa")
```

```
## Warning: package 'astsa' was built under R version 4.5.2
```

```r
library("xts")
```

```
## Warning: package 'xts' was built under R version 4.5.2

## Loading required package: zoo

## Warning: package 'zoo' was built under R version 4.5.2

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
Q <- (2^(1/3)*(pi)^(1/2)*exp(-4))/ (log(sin(2.75*pi)+cos(0.4*pi)))

print(Q)
```

```
## [1] 2.557122
```

```r
options(width = 40)
n = 1:10
sin(pi/n)
```

```
##  [1] 1.224606e-16 1.000000e+00
##  [3] 8.660254e-01 7.071068e-01
##  [5] 5.877853e-01 5.000000e-01
##  [7] 4.338837e-01 3.826834e-01
##  [9] 3.420201e-01 3.090170e-01
```

```r
round(sin(pi/n), 4)
```

```
##  [1] 0.0000 1.0000 0.8660 0.7071 0.5878
##  [6] 0.5000 0.4339 0.3827 0.3420 0.3090
```

```r
(1:10/5)
```

```
##  [1] 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8
## [10] 2.0
```

```r
(1:10) %% 3
```

```
##  [1] 1 2 0 1 2 0 1 2 0 1
```

```r
(1:9/10)
```

```
## [1] 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
```

```r
(0:8/10)
```

```
## [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
```

```r
j = seq(9,1, by = -8)
(j/10)
```

```
## [1] 0.9 0.1
```

```r
x <- c(1,2,3)
y <- c("hello", "string")
z <- c(TRUE, TRUE,FALSE, TRUE)
w <- c(1, TRUE , "bob")
```

```r
(x<-c(1,0,NA))
```

```
## [1]  1  0 NA
```

```r
y < x/0
```

```
## Warning in y < x/0: longer object
## length is not a multiple of shorter
## object length
```

```
## [1]  TRUE FALSE    NA
```

```r
y
```

```
## [1] "hello"  "string"
```

```r
x <- c(1,2,3,4);
y <- c(2,4, 6, 8);
z <- c(10, 20);
w <- c(7, 9, 11)

x+y
```

```
## [1]  3  6  9 12
```

```r
x+z
```

```
## [1] 11 22 13 24
```

```r
x*z
```

```
## [1] 10 40 30 80
```

```r
x-w
```

```
## Warning in x - w: longer object length
## is not a multiple of shorter object
## length

## [1] -6 -7 -8 -3
```

```r
ls()
```

```
## [1] "j" "n" "Q" "w" "x" "y" "z"
```

```r
#rm("varname")
data()
getwd()
```

```
## [1] "C:/Users/49603/Downloads/STAT 335 Time Series"
```

```r
#setwd("dir")
```

```r
write(uspop, file="uspop.txt", ncolumns=1)

read.table("uspop.txt")
```

```
##       V1
## 1   3.93
## 2   5.31
## 3   7.24
## 4   9.64
## 5  12.90
## 6  17.10
## 7  23.20
```

```
## 8   31.40
## 9   39.80
## 10  50.20
## 11  62.90
## 12  76.00
## 13  92.00
## 14 105.70
## 15 122.80
## 16 131.70
## 17 151.30
## 18 179.30
## 19 203.20
```

```
scan(file = "uspop.txt")
```

```
##  [1]   3.93   5.31   7.24   9.64  12.90
##  [6]  17.10  23.20  31.40  39.80  50.20
## [11]  62.90  76.00  92.00 105.70 122.80
## [16] 131.70 151.30 179.30 203.20
```

```
View("uspop.txt")
```

```
#View(swiss)

swiss[1]
```

```
##              Fertility
## Courtelary        80.2
## Delemont          83.1
## Franches-Mnt      92.5
## Moutier           85.8
## Neuveville        76.9
## Porrentruy        76.1
## Broye             83.8
## Glane             92.4
## Gruyere           82.4
## Sarine            82.9
## Veveyse           87.1
## Aigle             64.1
## Aubonne           66.9
## Avenches          68.9
## Cossonay          61.7
## Echallens         68.3
## Grandson          71.7
## Lausanne          55.7
## La Vallee         54.3
## Lavaux            65.1
## Morges            65.5
## Moudon            65.0
## Nyone             56.6
## Orbe              57.4
## Oron              72.5
```

```
## Payerne           74.2
## Paysd'enhaut      72.0
## Rolle             60.5
## Vevey             58.3
## Yverdon           65.4
## Conthey           75.5
## Entremont         69.3
## Herens            77.3
## Martigwy          70.5
## Monthey           79.4
## St Maurice        65.0
## Sierre            92.2
## Sion              79.3
## Boudry            70.4
## La Chauxdfnd      65.7
## Le Locle          72.7
## Neuchatel         64.4
## Val de Ruz        77.6
## ValdeTravers      67.6
## V. De Geneve      35.0
## Rive Droite       44.7
## Rive Gauche       42.8
```

```r
summary(swiss)
```

```
##    Fertility      Agriculture
##  Min.   :35.00   Min.   : 1.20
##  1st Qu.:64.70   1st Qu.:35.90
##  Median :70.40   Median :54.10
##  Mean   :70.14   Mean   :50.66
##  3rd Qu.:78.45   3rd Qu.:67.65
##  Max.   :92.50   Max.   :89.70
##   Examination      Education
##  Min.   : 3.00   Min.   : 1.00
##  1st Qu.:12.00   1st Qu.: 6.00
##  Median :16.00   Median : 8.00
##  Mean   :16.49   Mean   :10.98
##  3rd Qu.:22.00   3rd Qu.:12.00
##  Max.   :37.00   Max.   :53.00
##     Catholic       Infant.Mortality
##  Min.   :  2.150   Min.   :10.80
##  1st Qu.:  5.195   1st Qu.:18.15
##  Median : 15.140   Median :20.00
##  Mean   : 41.144   Mean   :19.94
##  3rd Qu.: 93.125   3rd Qu.:21.70
##  Max.   :100.000   Max.   :26.60
```

```r
write.table(swiss, "swiss.txt", sep=",")

read.table("swiss.txt", header = TRUE, sep=",")
```

```
##              Fertility Agriculture
## Courtelary        80.2        17.0
```

```
## Delemont          83.1        45.1
## Franches-Mnt      92.5        39.7
## Moutier           85.8        36.5
## Neuveville        76.9        43.5
## Porrentruy        76.1        35.3
## Broye             83.8        70.2
## Glane             92.4        67.8
## Gruyere           82.4        53.3
## Sarine            82.9        45.2
## Veveyse           87.1        64.5
## Aigle             64.1        62.0
## Aubonne           66.9        67.5
## Avenches          68.9        60.7
## Cossonay          61.7        69.3
## Echallens         68.3        72.6
## Grandson          71.7        34.0
## Lausanne          55.7        19.4
## La Vallee         54.3        15.2
## Lavaux            65.1        73.0
## Morges            65.5        59.8
## Moudon            65.0        55.1
## Nyone             56.6        50.9
## Orbe              57.4        54.1
## Oron              72.5        71.2
## Payerne           74.2        58.1
## Paysd'enhaut      72.0        63.5
## Rolle             60.5        60.8
## Vevey             58.3        26.8
## Yverdon           65.4        49.5
## Conthey           75.5        85.9
## Entremont         69.3        84.9
## Herens            77.3        89.7
## Martigwy          70.5        78.2
## Monthey           79.4        64.9
## St Maurice        65.0        75.9
## Sierre            92.2        84.6
## Sion              79.3        63.1
## Boudry            70.4        38.4
## La Chauxdfnd      65.7         7.7
## Le Locle          72.7        16.7
## Neuchatel         64.4        17.6
## Val de Ruz        77.6        37.6
## ValdeTravers      67.6        18.7
## V. De Geneve      35.0         1.2
## Rive Droite       44.7        46.6
## Rive Gauche       42.8        27.7
##              Examination Education
## Courtelary            15        12
## Delemont               6         9
## Franches-Mnt           5         5
## Moutier               12         7
## Neuveville            17        15
## Porrentruy             9         7
## Broye                 16         7
```

```
## Glane                 14         8
## Gruyere               12         7
## Sarine                16        13
## Veveyse               14         6
## Aigle                 21        12
## Aubonne               14         7
## Avenches              19        12
## Cossonay              22         5
## Echallens             18         2
## Grandson              17         8
## Lausanne              26        28
## La Vallee             31        20
## Lavaux                19         9
## Morges                22        10
## Moudon                14         3
## Nyone                 22        12
## Orbe                  20         6
## Oron                  12         1
## Payerne               14         8
## Paysd'enhaut           6         3
## Rolle                 16        10
## Vevey                 25        19
## Yverdon               15         8
## Conthey                3         2
## Entremont              7         6
## Herens                 5         2
## Martigwy              12         6
## Monthey                7         3
## St Maurice             9         9
## Sierre                 3         3
## Sion                  13        13
## Boudry                26        12
## La Chauxdfnd          29        11
## Le Locle              22        13
## Neuchatel             35        32
## Val de Ruz            15         7
## ValdeTravers          25         7
## V. De Geneve          37        53
## Rive Droite           16        29
## Rive Gauche           22        29
##              Catholic Infant.Mortality
## Courtelary       9.96             22.2
## Delemont        84.84             22.2
## Franches-Mnt    93.40             20.2
## Moutier         33.77             20.3
## Neuveville       5.16             20.6
## Porrentruy      90.57             26.6
## Broye           92.85             23.6
## Glane           97.16             24.9
## Gruyere         97.67             21.0
## Sarine          91.38             24.4
## Veveyse         98.61             24.5
## Aigle            8.52             16.5
## Aubonne          2.27             19.1
```

```
## Avenches         4.43             22.7
## Cossonay         2.82             18.7
## Echallens       24.20             21.2
## Grandson         3.30             20.0
## Lausanne        12.11             20.2
## La Vallee        2.15             10.8
## Lavaux           2.84             20.0
## Morges           5.23             18.0
## Moudon           4.52             22.4
## Nyone           15.14             16.7
## Orbe             4.20             15.3
## Oron             2.40             21.0
## Payerne          5.23             23.8
## Paysd'enhaut     2.56             18.0
## Rolle            7.72             16.3
## Vevey           18.46             20.9
## Yverdon          6.10             22.5
## Conthey         99.71             15.1
## Entremont       99.68             19.8
## Herens         100.00             18.3
## Martigwy        98.96             19.4
## Monthey         98.22             20.2
## St Maurice      99.06             17.8
## Sierre          99.46             16.3
## Sion            96.83             18.1
## Boudry           5.62             20.3
## La Chauxdfnd    13.79             20.5
## Le Locle        11.22             18.9
## Neuchatel       16.92             23.0
## Val de Ruz       4.97             20.0
## ValdeTravers     8.65             19.5
## V. De Geneve    42.34             18.0
## Rive Droite     50.43             18.2
## Rive Gauche     58.33             19.3
```